Healthcare_Energy_Questionnaire

Enumerator's name

-

- (●) Gian Teufen
- ( ) Till Häussner

Do you give consent to proceed with this survey?

- (●) Yes
- ( ) No

Questionnaire

» Facility Identifiers

GPS location

You can only proceed once the GPS' accuracy is below 500m.

latitude (x.y °)

_____

longitude (x.y °)

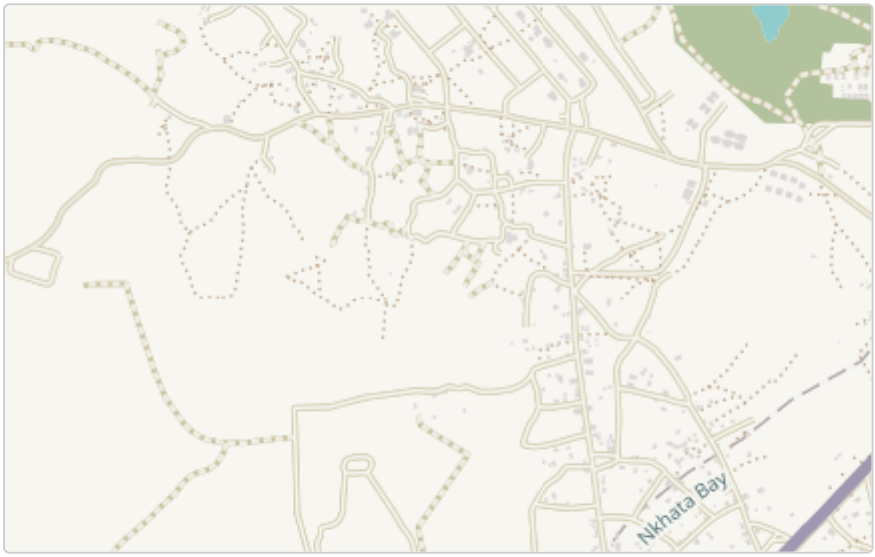
_____

altitude (m)

_____

accuracy (m)

_____



How far is the facility from the distribution lines of the grid? (in km)

Approximation of straight line distance to the nearest point of the grid.

_____

What is the name of the facility?

_____

Facility ID

In the order they are visited.

_____

In which region is the facility located?

- North
- Central
- South

In which district is the facility located?

- Balaka
- Blantyre
- Chikwawa
- Chiradzulu
- Chitipa
- Dedza
- Dowa
- Karonga
- Kasungu
- Likoma
- Lilongwe
- Machinga
- Mangochi
- Mchinji
- Mulanje
- Mwanza
- Mzimba
- Neno
- Nkhata Bay
- Nkhotakota
- Nsanje
- Ntcheu
- Ntchisi
- Phalombe
- Rumphi
- Salima
- Thyolo
- Zomba

What type of facility is this?

- ◯ National referral hospital
- ◯ District/Provincial hospital
- ◯ Health centre/clinic
- ◯ Health post
- ◯ Maternal/child health clinic
- ◯ Dispensary
- ◯ Rural Hospital
- ◯ Other (specify)

What type of facility is this?

_____

What is the managing authority of the facility?

- ◯ Government/Public
- ◯ NGO/Not-for-profit
- ◯ Private-for-profit
- ◯ Mission/faith-based
- ◯ Other (specify)

What is the managing authority of the facility?

_____

Is the facility located in an urban or rural area?

- ◯ Urban
- ◯ Rural

Is the facility outpatient only?

Excluding maternaty ward.

- ◯ Yes
- ◯ No
- ◯ Don't know

Is there a 24/7 maternity ward?

- ◯ Yes
- ◯ No
- ◯ Don't know

What is the catchment population size of the facility?

-99 if not known

_____

What is the catchment area of the facility?

-99 if not known

_____

On average, how many patients visit the facility per day?

- ◯ 0
- ◯ 1-10 patients
- ◯ 11-20 patients
- ◯ 21-50 patients
- ◯ 51-100 patients
- ◯ 101-200 patients
- ◯ >200 patients
- ◯ Don't know

How does the patient population vary over the year? Is there a busy season/which season is the busiest?

- ☐ Hot & Wet Season (November - April)
- ☐ Cool & Dry Season (May - August)
- ☐ Hot & Dry Season (September - October)
- ☐ No significant variation
- ☐ Don't know

Which days of the week is the facility open to patients?

- ☐ Monday
- ☐ Tuesday
- ☐ Wednesday
- ☐ Thursday
- ☐ Friday
- ☐ Saturday
- ☐ Sunday
- ☐ Don't know

Which days are the busiest per week on average?

- ☐ Monday
- ☐ Tuesday
- ☐ Wednesday
- ☐ Thursday
- ☐ Friday
- ☐ Saturday
- ☐ Sunday
- ☐ Don't know

On average, how many hours per day is this facility open?
Excluding off-schedule emergencies and maternity ward

_____

How long do outpatients stay in the clinic on average? (hours)

- ◯ <1 hour
- ◯ 1-2 hours
- ◯ 2-4 hours
- ◯ 4-6 hours
- ◯ >6 hours
- ◯ Don't know

How long is the average consultation?

- ◯ <5 minutes
- ◯ 5-15 minutes
- ◯ 15-30 minutes
- ◯ 30-60 minutes
- ◯ >60 minutes
- ◯ Don't know

How do patients get to the clinic?

- ☐ Foot
- ☐ Bicycle
- ☐ Car (private/taxi)
- ☐ Bus (public transport)
- ☐ Shuttle (facility-provided)
- ☐ Motorbike
- ☐ Oxcart
- ☐ Other (specify)

How do patients get to the clinic?

_____

» » Evening supply

On average, for how many hours during the evening (after 6 p.m.) and night is the facility open for services?
Excluding off-schedule emergencies and maternity ward

- ◯ Never open after 6 p.m
- ◯ Until 8 p.m
- ◯ Until 10 p.m.
- ◯ Until midnight
- ◯ All night long

On average, for how many hours during the evening (after 6 p.m.) and night is electricity available at the health facility?

- ◯ Less than 1 hour
- ◯ 1 to 2 hours
- ◯ 3 to 4 hours
- ◯ 5 to 8 hours
- ◯ 8 to 14 hours

On average, how many nights per month is a patient treated? (after 6 p.m.)

Including off-schedule emergencies, excluding maternity ward.

_____

On average, how many nights (after 6 p.m.) per month is a patient treated without adequate lighting?

_____

» Energy System

Does your facility have electricity from any source (e.g. grid, generator, solar, or other)?

- ◉ Yes
- ◯ No

» » Electricity System

What is the electricity used for in the facility?

- [x] Stand-alone electric medical devices/appliances (e.g. EPUI cold room, suction apparatus, refrigerator, etc.)
- [ ] Electric lighting (excluding flashlights)
- [ ] Staff housing
- [ ] Communications
- [ ] Device charging
- [ ] Water Pumps
- [ ] Heating, Ventilation, and Air Conditioning (HVAC)
- [ ] Security and Safety (security cameras, alarms, fire detection systems)
- [ ] Other (specify)

What is the electricity used for in the facility?

_____

If the facility has electricity, what is its main source? Note: This means the electricity source that you rely upon most of the time.

- (●) Central supply (national/community grid)
- ( ) Local mini-grid (i.e., the facility shares electricity supply, typically a renewable source or a generator, with other nearby buildings) Specify the supply modality if known.
- ( ) On-site hybrid solar/generator
- ( ) On-site solar system (except lanterns)
- ( ) On-site generator (with battery storage)
- ( ) On-site generator (no battery storage)
- ( ) Dedicated hydropower system
- ( ) Dedicated wind power system
- ( ) Other (Specify)

If the facility has electricity, what is its main source? Note: This means the electricity source that you rely upon most of the time.

_____

Other than the main or primary source, does the facility have a backup source of electricity?

- (●) Yes
- ( ) No
- ( ) Don't know

What is the secondary source of electricity?

- [x] No back-up source
- [x] Central supply electricity grid
- [x] Local mini-grid (i.e., the facility shares electricity supply, typically a renewable source or a generator, with other nearby buildings) Specify the supply modality if known.
- [x] On-site hybrid solar/generator
- [x] On-site solar system (except lanterns)
- [x] On-site generator (with battery storage)
- [x] On-site generator (no battery storage)
- [x] Dedicated hydropower system
- [x] Dedicated wind power system
- [x] Solar lanterns
- [ ] Other (Specify)

Other than the main or primary source, does the facility have a backup source of electricity? If so, what is the secondary source of electricity?

_____

More on this topic will later be recorded in the excel sheet.

Does the facility have an energy storage system like a rechargeable battery system?

- (●) Yes
- ( ) No
- ( ) Multiple
- ( ) Don't know

If you have both a primary and a secondary/backup source, do you have an energy management system to switch automatically?

- ( ) Yes, automatic shift, with fuel efficiency optimization (for example with the wind/solar)
- ( ) Yes, automatic shift; but no efficiency optimization
- ( ) No, a manual startup is needed

» » Alternative Lighting

Does the facility ever experience complete power loss?

- (●) Yes
- ( ) No
- ( ) Don't know

If the whole facility has no electricity, what is do you use as a lighting source?

- [ ] No source of lighting
- [ ] Kerosene lamps
- [ ] Candles
- [ ] Battery-operated lamps/flashlights
- [ ] Solar lanterns
- [ ] Headlamps
- [x] Other (Specify)

If the whole facility has no electricity, what is its backup lighting source?

» » Generator Functionality

Is the generator functional?

- (●) Yes
- ( ) No
- ( ) Don't know

Does the generator have fuel available today?

- [ ] Yes
- [ ] Petrol
- [ ] Diesel
- [ ] No
- [ ] Don't know

More on this topic will later be recorded in the excel sheet.

» » Solar System Functionality

More on this topic will later be recorded in the excel sheet.

» » Grid Reliability

Is the grid functional?

- (●) Yes
- ( ) No
- ( ) Don't know

In recent times, how often was there a power outage?

- ( ) Multiple times a day
- ( ) Every day
- ( ) Every 2 days
- ( ) Every 3-4 to days
- ( ) Every week
- ( ) Every two weeks
- ( ) Every month
- ( ) Rarely
- ( ) Never

In the case of a grid power outage, how many hours does it last on average?

-99 if not known

_____

When are supply interruptions most common?

- ( ) Daytime (before dark)
- ( ) Evening (after dark to midnight)
- ( ) Overnight (midnight to 8 a.m.)
- ( ) Don't know

» » Quality

Does the facility usually face a problem of low voltage or voltage fluctuations from the main source?

- (●) Yes
- ( ) No
- ( ) Don't know

Have voltage fluctuations damaged equipment in the past year?

- ( ) No
- ( ) Some equipment damaged
- ( ) Significant equipment damaged

» » Financial Sustainability

If you use electricity from the grid, did you have sufficient funds available for paying your electricity bills in the last two months?

- ( ) Yes
- ( ) No
- (●) Don't know
- ( ) Not applicable

If you use electricity from the grid, who pays the electricity bill for the health clinic?

- [ ] MoH/DHO with additional budget
- [ ] Donor
- [ ] Health facility baseline budget
- [ ] Health facility comittee
- [ ] Staff for their own housing
- [ ] Patients
- [ ] Don't know
- [x] Other (Specify)
- [ ] No one

If you use electricity from the grid, who pays the electricity bill for the health clinic?

_____

Over the last two months, did you have sufficient funds available for fuel for the stand-alone generator to operate at all times when it was needed?

- ( ) Yes
- ( ) No
- (●) Don't know

Who pays for generator fuel?

- [ ] MoH/DHO with additional budget
- [ ] Donor
- [ ] Health facility baseline budget
- [ ] Health facility comittee
- [ ] Staff
- [ ] Patients
- [ ] Don't know
- [x] Other (Specify)
- [ ] No one

Who pays for generator fuel?

Over the last year, did you have sufficient funds available to keep the solar panels functional?

- ( ) Yes
- ( ) No
- ( ) Don't know

If you use a solar energy-based electricity solution, who pays for light bulbs and solar parts replacement?

- [ ] MoH/DHO with additional budget
- [ ] Donor
- [ ] Health facility baseline budget
- [ ] Health facility comittee
- [ ] Staff
- [ ] Patients
- [ ] Don't know
- [x] Other (Specify)
- [ ] No one

If you use a solar energy-based electricity solution, who pays for light bulbs and solar parts replacement?

» Origins, Operations and Maintenance

» » Maintenance (preventive)

Is preventive maintenance carried out for any of the energy systems?

Preventive maintenance means the service is carried out even when there is no problem with the system.

- ( ) Routinely
- ( ) Sometimes but not routinely
- (•) Never

» » Repair (corrective)

Is there a designated staff member trained in managing or troubleshooting any part of the energy system?

- ○ Yes
- ○ No
- ○ Partially
- ○ Don't know

Elaborate on what the training consisted of.

» » Suitability and Future Needs

Is the current electricity system sufficient for your current needs?

- ◉ Yes
- ○ No
- ○ Sometimes
- ○ Don't know

Do you expect your electricity needs to increase in the next 12 months?

- ○ Yes
- ○ No
- ○ Don't know

Are there plans to upgrade or expand the electricity system?

- ○ Yes
- ○ No
- ○ Don't know

What kind of upgrade is planned?

- ☐ Central supply (national/community grid)
- ☐ Local mini-grid
- ☐ On-site hybrid solar/generator
- ☐ On-site generator (with battery storage)
- ☐ On-site generator (no battery storage)
- ☐ On-site solar system (except lanterns)
- ☐ Dedicated hydropower system
- ☐ Dedicated wind power system
- ☐ Rechargeable battery system
- ☐ Solar lanterns
- ☐ Other (Specify)

What kind of upgrade is planned?

» Climate Change Impact

» » Facility Impact

Have you observed any changes in weather or climate over the past few years?

- [ ] No
- [ ] Increased temperatures
- [ ] Heatwaves
- [ ] More frequent storms
- [ ] Flooding
- [ ] Longer dry periods
- [x] Other (specify)

Have you observed any changes in weather or climate over the past few years?

Have changes in climate or weather events affected the operations at your facility?

- (●) Yes, significantly
- ( ) Yes, somewhat
- ( ) No
- ( ) Don't know

What parts of facility operations have been most affected by extreme weather events?

- [ ] Temporary increases in patients due to more malaria in the rainy season
- [ ] Temporary increases in patients due to malnutrition in long dry seasons
- [ ] Temporary increases in patients in cold, dry seasons due to dust in the air, causing cough
- [ ] Temporary decreases in facility accessibility due to floods
- [ ] Temporary decreases in facility accessibility due to wild fires
- [ ] Temporary decreases in facility accessibility due to storms/cyclones
- [ ] Injuries because of storms/cyclones/lightnings
- [ ] Damage to infrastructure or energy systems or equipment because of storms/cyclones/lightnings
- [ ] Grid is less reliable during dry seasons
- [ ] Decreased water availability
- [ ] Migration to/from the region due to weather/environment
- [ ] People not seeking care due to power outages
- [ ] Delayed care-seeking due to transportation issues
- [ ] Reduced ability to refer patients to other facilities due to weather
- [ ] Rooms too hot so doors have to be opened, affects privacy
- [ ] Don't know
- [x] Other (specify)

What parts of facility operations have been most affected by extreme weather events?

_____

» » Climate Change Perception

Based on your current knowledge, what or who is the cause of climate change?

- ( ) Human activities (e.g. burning fossil fuels, deforestation)
- ( ) Natural processes (e.g. volcanic activity, changes in solar radiation)
- ( ) Both human and natural causes
- ( ) Climate change is not happening
- ( ) I don't know
- (•) Other (specify)

Based on your current knowledge, what or who is the cause of climate change?

_____

In your opinion, is climate change affecting your community or the way health services are delivered?

- ( ) Yes, significantly
- ( ) Yes, to some extent
- ( ) No
- (•) Don't know

» Amenities and Equipment

» » Basic Amenities

Is there a water source available?

- (●) Yes
- ( ) No

What does the water source of the facility look like?

- [ ] Piped
- [ ] Public tap
- [ ] Standpipe
- [ ] Tubewell/Borehole
- [ ] Protected Dug Well
- [ ] Protected Spring
- [ ] Rain Water
- [ ] None

Where is the water source located?

- ( ) Inside facility
- ( ) Outside, within grounds
- ( ) Outside, external
- ( ) Other (specify)

Where is the water source located?

_____

Is there a separate, private consultation room available?

- ( ) Yes
- ( ) No

Can a normal conversation be held in the consultation room without being overheard?

- ( ) Yes
- ( ) No

Is there a toilet/latrine available for clients?

- (●) Yes
- ( ) No

What does the toilet/latrine look like?

- [ ] Flush/pour flush to piped sewer system or septic tank or pit latrine
- [ ] pit latrine (ventilated improved pit (VIP) or other) with slab
- [ ] composting toilet
- [ ] None

Are there adequate sanitation facilities available on premises for clients?

- ○ Yes
- ● No

What functioning communication equipment is available?

- ☐ Private cell phone (reimbursed by facilitiy)
- ☐ Private cell phone
- ☐ Facility cell phone
- ☐ Other phone
- ☐ SW radio

Is there a functioning facility computer available?

- ○ Yes
- ○ No

Does the computer have access to working internet? (day of survey)

- ○ Yes
- ○ No

Is there an emergency transportation vehicle available for clients?

- ○ Yes; facility-owned, routinely available, with fuel
- ○ Yes; access to vehicle near facility, routinely available, with fuel
- ○ No

» » Basic Equipment

Adult Scale

- ○ Yes
- ○ No

Child Scale (250g steps min)

- ○ Yes
- ○ No

Thermometer

- ○ Yes
- ○ No

Stethoscope

- ○ Yes
- ○ No

Blood pressure apparatus

- ○ Yes
- ○ No

Light source (Spotlight/flashlight for examinations)

- ◯ Yes
- ◯ No

» » Standard precautions for infection prevention

How are sharps disposed of?

- ◯ Incinerator (functioning & fuel available)
- ◯ Open burning in unprotected area
- ◯ Open burning in protected area
- ◯ Dump without burning in protected area
- ◯ Remove offsite with protected storage
- ◯ Other (specify)
- ◯ None
- ◯ Open burning in protected area

How are sharps disposed of?

Are unprotected sharps observed on site?

- ◯ Yes
- ◯ No

How is infectious waste disposed of?

- ◯ Incinerator (functioning & fuel available)
- ◯ Open burning in unprotected area
- ◯ Dump without burning in protected area
- ◯ Remove offsite with protected storage
- ◯ Other (specify)
- ◯ None

How is infectious waste disposed of?

Is unprotected infectious waste observed on site?

- ◯ Yes
- ◯ No

Is storage for sharps observed on site?

- ◯ Yes
- ◯ No

What does the storage for sharps look like?

- [ ] Puncture-resistant
- [ ] Rigid
- [ ] Leak-resistant
- [ ] Container to hold used sharps safely during collection, disposal and distruction
- [ ] Material: plastic
- [ ] Material: metal
- [ ] Material: cardboard
- [ ] Closeable
- [ ] For sharps of all standard sizes
- [ ] Clearly marked with intl. Biohazard warning no less than 50mm in diameter, printed in black or red on each of the front an back faces of the box
- [ ] Other (specify)
- [ ] None

What does the storage for sharps look like?

_____

Where was the storage for sharps observed?

- [ ] General OPD
- [ ] HIV testing area
- [ ] Surgery area
- [ ] Other (specify)
- [ ] None

Where was the storage for sharps observed?

_____

Is there appropriate storage for infectious waste? (Waste receptacle (pedal bin) with lid and plastic bin liner)

- ( ) Yes
- ( ) No

Is there disinfectant available anywhere in the facility? (Chlorine-based or other country specific used for environmental disinfection)

- ( ) Yes
- ( ) No

Are there Single use —standard disposable or auto-disable syringes available anywhere in the facility?

- ( ) Yes
- ( ) No

Where is Soap and running water or alcohol based hand rub available in the facilitiy?

- [ ] General OPD
- [ ] HIV testing area
- [ ] Basic obstetric area
- [ ] Newborn care area
- [ ] Surgery area
- [ ] Other (specify)
- [ ] None

Where is Soap and running water or alcohol based hand rub available in the facilitiy?

_____

Where are Latex gloves available in the facilitiy? (If equivalent non latex gloves are available this is acceptable.)

- [ ] General OPD
- [ ] HIV testing area
- [ ] Basic obstetric area
- [ ] Newborn care area
- [ ] Surgery area
- [ ] Other (specify)
- [ ] None

Where are Latex gloves available in the facilitiy? (If equivalent non latex gloves are available this is acceptable.)

_____

Are there guidelines for standard precautions available anywhere in the facility?

- ( ) Yes
- ( ) No

» » Diagnostic capacity

Haemoglobin (This may include colorimeter OR haemoglobinometer OR hemocue.)

- ( ) Yes
- ( ) No

Blood glucose (Glucometer and glucometer test strips)

- ( ) Yes
- ( ) No

Malaria diagnostic capacity (RDT kit or smear with microscope, slides, and Wright Giemsa stain)

- ( ) Yes
- ( ) No

Urine dipstick- protein (Dipsticks for urine protein (with valid expiration date))

- ◯ Yes
- ◯ No

Urine dipstick- glucose (Dipsticks for urine glucose (with valid expiration date))

- ◯ Yes
- ◯ No

HIV diagnostic capacity (RDT kit or ELISA test with ELISA washer, ELISA reader, incubator, specific assay kit)

- ◯ Yes
- ◯ No

Syphilis rapid test (RDT kit)

- ◯ Yes
- ◯ No

Urine test for pregnancy (RDT kit)

- ◯ Yes
- ◯ No

» » Essential Medicines

Amlodipine tablet or alternative calcium channel blocker

- ◯ Yes
- ◯ No

Amoxicillin syrup/suspension or dispersible tablet

- ◯ Yes
- ◯ No

Amoxicillin tablet (Respiratory antibiotic)

- ◯ Yes
- ◯ No

Ampicillin powder for injection

- ◯ Yes
- ◯ No

Aspirin cap/tab

- ◯ Yes
- ◯ No

Beclometasone inhaler

- ◯ Yes
- ◯ No

Beta blocker (e.g.bisoprolol, metoprolol, carvedilol, atenolol)

- ◯ Yes
- ◯ No

Carbamazepine tablet

- ◯ Yes
- ◯ No

Ceftriaxone injection (2nd line injectable antibiotic)

- ◯ Yes
- ◯ No

Diazepam injection

- ◯ Yes
- ◯ No

Enalapril tablet or alternative ACE inhibitor e.g. lisinopril, ramipril, perindopril

- ◯ Yes
- ◯ No

Fluoxetine tablet

- ◯ Yes
- ◯ No

Gentamicin injection

- ◯ Yes
- ◯ No

Glibenclamide tablet (Oral treatment type 2 diabetes)

- ◯ Yes
- ◯ No

Haloperidol tablet

- ◯ Yes
- ◯ No

Insulin regular injection

- ◯ Yes
- ◯ No

Magnesium sulphate injectable

- ◯ Yes
- ◯ No

Metformin tablet

- ◯ Yes
- ◯ No

Omeprazole tablet or alternative such as pantoprazole, rabeprazole (Gastro-esophogeal reflux)

- ◯ Yes
- ◯ No

Oral rehydration solution

- ◯ Yes
- ◯ No

Oxytocin injection

- ◯ Yes
- ◯ No

Salbutamol inhaler (Chronic asthma attacks)

- ◯ Yes
- ◯ No

Simvastatin tablet or other statin e.g. atorvastatin, pravastatin, fluvastatin (High cholesterol)

- ◯ Yes
- ◯ No

Thiazide (e.g. hydrochlorothiazide)

- ◯ Yes
- ◯ No

Zinc sulphate tablets, dispersible tablets or syrup

- ◯ Yes
- ◯ No

Thank you for your time. The survey is now finished.This space allows you to add anything you think should be mentioned.

_____